COMP4141 Slides Week 4

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Admin

Tests

- Marks will be on formatif
- Test is closed book (just to clarify)
- You will not be getting you sheet back but you can ask if you had questions
- For week 3 (last week) some lost 1 mark for having a solution that doesn't use the ϵ-NFA construction like the question asks for.
- The **week-5** (next week) in-class assessment will be to use the context-free pumping lemma to prove some language to be non-context-free.

Marking

For week 1:

- The general patterns were most did not reference/justify why $|aa| = 2$ or $|x.y| = |x| + |y|$.
- Some used an alternative definition such as ax instead of xa.
- Some used induction on the length of the string directly instead of a structural induction.

For week 2:

- Pass and Credit tasks were generally well done by those who attempted them.
- For the Distinction task, many students had overly long or complicated proofs. Try to use structured induction (I recommend using the tutorial solution as a guide).
- HD task was correct on the first attempt (the counterexample was simple).

Starts on Marking Moving forwards

- This will have:
  - **0 star:** not acceptable, completely wrong.
  - **1 star:** on the right track, but not sufficient and details missing.
  - **2 stars:** almost there, but somewhat incomplete.
  - **3 starts:** all correct.

More on Marking

- Consider lowering your target grade (or raising it) if necessary!

Marking

Apologies for any confusion before

- Those who submit before the due date get feedback. The feedback comes in one week from me, and then you will have another week to take the feedback and revise. Together, this is a three week process, and two weeks after the first due date.
- You can still submit after the due date (first due date), and before the deadline (final due date).

Properties of CFLs

Pumping Lemma for CFLs

Pumping Lemma

If $L \subseteq \Sigma^*$ is context–free, then there exists $p \in \mathbb{N}$ (the pumping length) such that if $w \in L$ with $|w| \geq p$, then w can be written as

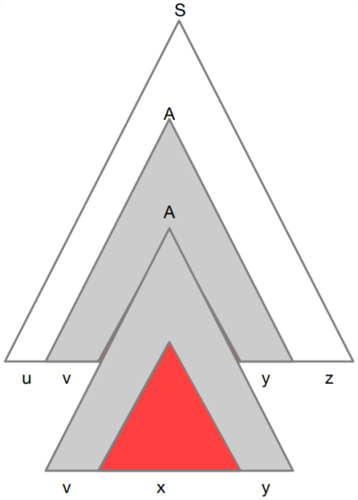
$$w = uvxyz$$

satisfying

- $|vy| > 0$,
- $|vxy| \leq p$,
- $uv^i xy^i z \in L$ for all $i \in \mathbb{N}$.

Pase Tree Intuition

In a branch we have to reach a variable twice for a large enough word.



CFL Algorithms

Chomsky Normal Form

Chomsky Normal Form

A context-free grammar is in **Chomsky normal form (CNF)** if every production is of one of the forms

$$S \longrightarrow \epsilon, \qquad A \longrightarrow BC, \qquad A \longrightarrow a,$$

where B, C are variables (and neither is S) and $a \in \Sigma$.

Theorem

Every context-free language is generated by a CFG in Chomsky normal form.

Converting to Chomsky Normal Form

Algorithm

1. **Add a new start variable.** Introduce a fresh start symbol S_0 with production $S_0 \longrightarrow S$. Remove ϵ-productions (except possibly from S_0).
2. **Remove terminals from long RHS.** Replace terminals appearing with variables by new variables X_a and add productions $X_a \longrightarrow a$.
3. **Remove unit productions** ($A \longrightarrow B$). Replace them by the productions of B.
4. **Reduce RHS to pairs of variables.** Replace productions of the form

$$A \longrightarrow B_1 B_2 \cdots B_n \quad (n > 2)$$

with

$$A \longrightarrow B_1 C_1,\ C_1 \longrightarrow B_2 C_2,\ \ldots,\ C_{n-2} \longrightarrow B_{n-1} B_n$$

using fresh variables C_i.

CYK Algorithm

CYK Algorithm

Let $G = (V, \Sigma, P, S)$ be a CFG in Chomsky normal form and let $w = a_1 \cdots a_n$.

1. **Base case.** For $i = 1, \ldots, n$ set
$$N_{ii} = \{A \in V : A \longrightarrow a_i \in P\}.$$
2. **Build larger substrings.** For $d = 1, \ldots, n-1$ and $i = 1, \ldots, n-d$, let $j = i + d$ and compute
$$N_{ij} = \{A \in V : A \longrightarrow BC \in P,\ B \in N_{ik},\ C \in N_{k+1,j} \text{ for some } i \leq k < j\}.$$
3. **Accept if $S \in N_{1n}$.**

W4 P1

Problem 1

Show that the following languages are not context-free:

(a) $\{0^n 1^n 0^n 1^n : n \in \mathbb{N}\}$

Problem 1

Show that the following languages are not context-free:

(b) $\{w\#t : w, t \in \{0,1\}^*$ and w is a substring of $t\}$

Problem 1

Show that the following languages are not context-free:

(c) $\{0^{2^n} : n \in \mathbb{N}\}$

W4 P2

Problem 2 (a)

Solve the following problems by first converting the grammar to Chomsky Normal Form, if necessary, and then applying the CYK algorithm.

(a) Let G_1 be the grammar $(\{S, B\}, \{a, b, c\}, R, S)$ where R consists of the following rules:

$$\begin{aligned} S &\rightarrow aSc \mid ac \mid B \\ B &\rightarrow \epsilon \mid Bb \end{aligned}$$

Decide if $abbc \in L(G_1)$

Problem 2 (a) CYK Table

a	b	b	c

Problem 2 (b)

Solve the following problems by first converting the grammar to Chomsky Normal Form, if necessary, and then applying the CYK algorithm.

(b) Let G_2 be the grammar $(\{S, A, B, C\}, \{a, b, c\}, R, S)$ where R consists of the following rules:

$$\begin{array}{lcl} S & \to & AB \\ A & \to & CC \mid a \mid c \\ B & \to & BC \mid b \\ C & \to & CB \mid BA \mid c \end{array}$$

Decide if $abbc \in L(G_2)$

Problem 2 (b) CYK Table

a	b	b	c

Problem 2 (c)

Solve the following problems by first converting the grammar to Chomsky Normal Form, if necessary, and then applying the CYK algorithm.

(c) Let G_3 be the grammar $(\{S, A, B, C\}, \{a, b, c\}, R, S)$ where R consists of the following rules:

$$
\begin{aligned}
S &\longrightarrow ASB \mid \epsilon \\
A &\longrightarrow C \mid BB \mid a \\
B &\longrightarrow BCC \mid b \mid \epsilon \\
C &\longrightarrow CBA \mid BAC \mid c
\end{aligned}
$$

Decide if $abbc \in L(G_3)$

Problem 2 (c) CYK Table

a	b	b	c

Feedback

Please let me know how I can improve for next week here!

COMP4141 26T1 Feedback Form
(For Brendan Mabbutt's Tutorials)